Questionnaire survey on conflicts between nature restoration and recreational use in parks and green spaces

This survey asks park management organizations and government officials about their views on the management and operation of urban parks and other green spaces where nature restoration is undertaken, and about measures and practical approaches to balancing nature restoration and recreational use.

We would appreciate a response from someone familiar with the actual conditions of facility development and management and natural environment conservation in the park (埼玉県生態系保護協会; Saitama Prefecture Ecosystem Conservation Association).

The questions concern the effects of various forms of park use by the general public on nature restoration activities undertaken within the park. In this questionnaire, nature restoration and recreational use are defined as follows:

* Nature restoration: activities undertaken within the park to conserve and restore the habitats of diverse organisms.

* Recreational use: park users' use of the various facilities within the park for recreation and other activities.

The questionnaire comprises 12 pages, including the cover and question pages.

After completing the questionnaire, please place it in the envelope, seal the envelope, and return it by ____ (month) ____ (day). If you have any questions, please contact the people listed below.

We appreciate that you are busy and would be grateful for your understanding of the purpose of this survey and your cooperation.

Survey details

(1) Student conducting the survey

Junying Chen

Third-year doctoral student, Science and Engineering program

Graduate School of Science and Engineering, Saitama University

(2) Contact information

Junying Chen (survey coordinator)

Email: chin.s.889@ms.saitama-u.ac.jp

Kiyotaka Fukahori (survey supervisor; Associate Professor, Graduate School of Science and Engineering, Saitama University)

Tel.: 048-858-9549

Email: fukahori@mail.saitama-u.ac.jp

Guidance for completing the questionnaire

This questionnaire is mainly intended for government departments, organizations, designated managers, experts/advisors, and others involved in managing nature in parks. It asks about measures for nature restoration to enhance biodiversity, sustainable park use, and urban beautification.

How to answer the questions (example)

Please mark the applicable response options clearly, as shown below.

Example of a completed response

Restriction status: indicate whether restrictions apply.

Degree of restriction: check one or two applicable options.

Activities subject to prohibitions or restrictions	Restrictions apply	No restrictions	Completely prohibited	Permitted only in certain areas	Permitted only during events
(1) Taking home organisms such as plants and insects	☐	☐	☑	☐	☑
(2) Bringing pets into the park	☐	☐	☐	☐	☑
(3) Entering protected areas important as habitats for organisms	☐	☐	☐	☐	☑
(4) Using tents (including sun shelters), tarps, etc.	☐	☐	☐	☐	☑
(5) Playing music or using sound equipment	☐	☐	☐	☐	☑
(6) Barbecuing or other eating and drinking activities involving fire	☐	☑	☐	☐	☐
(7) Using nighttime lighting in habitats of insects and other organisms	☐	☐	☐	☐	☐
(8) Sports and games, such as ball games	☐	☐	☑	☐	☐
(9) Fishing	☐	☐	☐	☐	☐
(10) Water-based recreation, such as swimming and boating	☐	☐	☐	☐	☐

If there are any other rules for park users intended to conserve the natural environment, please describe them in the space provided.

Response:

If you have information sheets, brochures, or other documents introducing the park, we would appreciate it if you could enclose them with the completed questionnaire in the return envelope.

1. Park-use rules

Q1. We would like to ask about the rules for using the park.

Please indicate the rules applied in day-to-day park operations, rather than answering solely on the basis of formal rules such as park ordinances. (If an activity is tolerated in practice, please select "No restrictions.")

Restriction status: indicate whether restrictions apply.

Degree of restriction: check one or two applicable options.

Activities subject to prohibitions or restrictions	Restrictions apply	No restrictions	Completely prohibited	Permitted only in certain areas	Permitted only during events
(1) Taking home organisms such as plants and insects	☐	☐	☐	☐	☐
(2) Bringing pets into the park	☐	☐	☐	☐	☐
(3) Entering protected areas important as habitats for organisms	☐	☐	☐	☐	☐
(4) Using tents (including sun shelters), tarps, etc.	☐	☐	☐	☐	☐
(5) Playing music or using sound equipment	☐	☐	☐	☐	☐
(6) Barbecuing or other eating and drinking activities involving fire	☐	☐	☐	☐	☐
(7) Using nighttime lighting in habitats of insects and other organisms	☐	☐	☐	☐	☐
(8) Sports and games, such as ball games	☐	☐	☐	☐	☐
(9) Fishing	☐	☐	☐	☐	☐
(10) Water-based recreation, such as swimming and boating	☐	☐	☐	☐	☐

If there are any other rules for park users intended to conserve the natural environment, please describe them in the space provided.

Response:

2. Organization of activities

Q2. We would like to ask about the current implementation of natural environment conservation activities in this park/green space.

Please check the applicable response options.

Natural environment conservation activities	Currently implemented	Recognized as a management issue or as requiring improvement, regardless of whether the activity is implemented
(1) Conservation deliberations: consideration of approaches to nature conservation in park management (e.g., review meetings and consultative councils).	☐ Yes ☐ No	☐ Yes ☐ No
(2) Habitat management: creating suitable habitats for organisms and managing environmental conditions (e.g., mowing, thinning, and improvement of waterside environments).	☐ Yes ☐ No	☐ Yes ☐ No
(3) Protection of wild plants and animals: conservation activities directed at species native to the region, such as protecting or translocating rare or valuable species and wild plants and animals.	☐ Yes ☐ No	☐ Yes ☐ No
(4) Removal of non-native species: activities to remove or control non-native species.	☐ Yes ☐ No	☐ Yes ☐ No
(5) Environmental surveys: surveys and monitoring of plants and animals, or hands-on environmental survey activities.	☐ Yes ☐ No	☐ Yes ☐ No
(6) Visitor information/displays: outreach activities, including environmental education through displays and information signs within the park.	☐ Yes ☐ No	☐ Yes ☐ No
(7) Environmental education: instructor-led environmental education events, such as seminars and lectures.	☐ Yes ☐ No	☐ Yes ☐ No
(8) Native species planting: except for commemorative or donated tree plantings and flowerbeds, tree planting and similar planting within the park are based mainly on native species, including local species.	☐ Yes ☐ No	☐ Yes ☐ No

Q3. We would like to ask about the arrangements for implementing the natural environment conservation activities reported in Q2 (hereafter, "activities").

Are the activities in Q2 undertaken by the local government itself or with the cooperation of organizations and other groups? Please check the applicable implementing bodies and enter the activity numbers.

Example of a completed response

(1) Please check the organizations or groups involved in implementing the activities.

(2) For the activities currently implemented in Q2 (see the activity reference list), enter the numbers (1)-(8) in the response column for the applicable implementing bodies.

* If no activity number applies, you may leave the response space blank.

Implementing bodies	Activity numbers
☐ Local governments and government agencies	
☑ Designated managers	(1), (5)
☐ NPOs/citizen groups other than designated managers	
☐ Groups formed by members of the public specifically for the park	
☑ General park users and nearby residents	(1)
☑ Educational and research institutions, such as schools	(1)
☑ Experts/advisors	(1), (7)
☐ Other	

Activity reference list (items from Q2)

(1) Conservation deliberations: consideration of approaches to nature conservation in park management (e.g., review meetings and consultative councils).

(2) Habitat management: creating suitable habitats for organisms and managing environmental conditions (e.g., mowing, thinning, and improvement of waterside environments).

(3) Protection of wild plants and animals: conservation activities directed at species native to the region, such as protecting or translocating rare or valuable species and wild plants and animals.

(4) Removal of non-native species: activities to remove or control non-native species.

(5) Environmental surveys: surveys and monitoring of plants and animals, or hands-on environmental survey activities.

(6) Visitor information/displays: outreach activities, including environmental education through displays and information signs within the park.

(7) Environmental education: instructor-led environmental education events, such as seminars and lectures.

(8) Native species planting: except for commemorative or donated tree plantings and flowerbeds, tree planting and similar planting within the park are based mainly on native species, including local species.

↓

If there are several activities, list their numbers as shown in the example. If several organizations or groups jointly undertake an activity, enter the same activity number in the response space for each applicable organization or group.

Please provide your responses below.

(1) Please check the organizations or groups involved in implementing the activities.

(2) For the activities currently implemented in Q2 (see the activity reference list), enter the numbers (1)-(8) in the response column for the applicable implementing bodies.

* If no activity number applies, you may leave the response space blank.

Implementing bodies	Activity numbers
☐ Local governments and government agencies	
☐ Designated managers	
☐ NPOs/citizen groups other than designated managers	
☐ Groups formed by members of the public specifically for the park	
☐ General park users and nearby residents	
☐ Educational and research institutions, such as schools	
☐ Experts/advisors	
☐ Other	

Activity reference list (items from Q2)

(1) Conservation deliberations: consideration of approaches to nature conservation in park management (e.g., review meetings and consultative councils).

(2) Habitat management: creating suitable habitats for organisms and managing environmental conditions (e.g., mowing, thinning, and improvement of waterside environments).

(3) Protection of wild plants and animals: conservation activities directed at species native to the region, such as protecting or translocating rare or valuable species and wild plants and animals.

(4) Removal of non-native species: activities to remove or control non-native species.

(5) Environmental surveys: surveys and monitoring of plants and animals, or hands-on environmental survey activities.

(6) Visitor information/displays: outreach activities, including environmental education through displays and information signs within the park.

(7) Environmental education: instructor-led environmental education events, such as seminars and lectures.

(8) Native species planting: except for commemorative or donated tree plantings and flowerbeds, tree planting and similar planting within the park are based mainly on native species, including local species.

Q4. We would like to ask about current measures to improve the convenience of recreational use in this park/green space and your perceptions of their adverse impacts.

Items relating to the convenience of recreational use	**If an item in the left-hand column is currently implemented, please check the box in the upper section. Then also check the appropriate box in the lower section regarding adverse environmental impacts resulting from that measure.**
(1) To protect park users' safety, hazardous species, such as bees/wasps and poisonous or venomous insects, and their nests are removed when found.	☐ **Currently implemented** Regarding adverse impacts of (1) on the environment and organisms: ☐ There are no problems, and no countermeasures have been taken. ☐ Impacts have been mitigated through countermeasures or other adjustments. ☐ No countermeasures have been taken; impacts exist but are tolerated. ☐ No countermeasures have been taken, but the need for improvement is recognized.
(2) Enjoyable events are held that showcase the park's natural attractions or make use of its natural features, and information is provided through interpretive signs and publicity materials.	☐ **Currently implemented** Regarding adverse impacts of (2) on the environment and organisms: ☐ There are no problems, and no countermeasures have been taken. ☐ Impacts have been mitigated through countermeasures or other adjustments. ☐ No countermeasures have been taken; impacts exist but are tolerated. ☐ No countermeasures have been taken, but the need for improvement is recognized.
(3) Places or facilities are available for eating and drinking involving the use of fire, such as barbecuing.	☐ **Currently provided** Regarding adverse impacts of (3) on the environment and organisms: ☐ There are no problems, and no countermeasures have been taken. ☐ Impacts have been mitigated through countermeasures or other adjustments. ☐ No countermeasures have been taken; impacts exist but are tolerated. ☐ No countermeasures have been taken, but the need for improvement is recognized.
(4) Main park paths are paved with non-natural materials to provide barrier-free access and accommodate wheelchair users.	☐ **Currently provided** Regarding adverse impacts of (4) on the environment and organisms: ☐ There are no problems, and no countermeasures have been taken. ☐ Impacts have been mitigated through countermeasures or other adjustments. ☐ No countermeasures have been taken; impacts exist but are tolerated. ☐ No countermeasures have been taken, but the need for improvement is recognized.
(5) Trash bins are installed in the park.	☐ **Currently installed** Regarding adverse impacts of (5) on the environment and organisms: ☐ There are no problems, and no countermeasures have been taken. ☐ Impacts have been mitigated through countermeasures or other adjustments. ☐ No countermeasures have been taken; impacts exist but are tolerated. ☐ No countermeasures have been taken, but the need for improvement is recognized.

(6) Water-based recreation, such as boating, fishing, and playing in the water, is available at ponds and other waterside areas.	☐ **Currently available for use** Regarding adverse impacts of (6) on the environment and organisms: ☐ There are no problems, and no countermeasures have been taken. ☐ Impacts have been mitigated through countermeasures or other adjustments. ☐ No countermeasures have been taken; impacts exist but are tolerated. ☐ No countermeasures have been taken, but the need for improvement is recognized.
(7) Insects may be collected freely.	☐ **Currently permitted** Regarding adverse impacts of (7) on the environment and organisms: ☐ There are no problems, and no countermeasures have been taken. ☐ Impacts have been mitigated through countermeasures or other adjustments. ☐ No countermeasures have been taken; impacts exist but are tolerated. ☐ No countermeasures have been taken, but the need for improvement is recognized.
(8) Organisms from outside the park (e.g., fireflies and fish) may be released into the park, or animals in the park may be fed.	☐ **Currently permitted** Regarding adverse impacts of (8) on the environment and organisms: ☐ There are no problems, and no countermeasures have been taken. ☐ Impacts have been mitigated through countermeasures or other adjustments. ☐ No countermeasures have been taken; impacts exist but are tolerated. ☐ No countermeasures have been taken, but the need for improvement is recognized.
(9) Park users may bring pets into the park, or a dog run has been provided.	☐ **Currently available for use** Regarding adverse impacts of (9) on the environment and organisms: ☐ There are no problems, and no countermeasures have been taken. ☐ Impacts have been mitigated through countermeasures or other adjustments. ☐ No countermeasures have been taken; impacts exist but are tolerated. ☐ No countermeasures have been taken, but the need for improvement is recognized.
(10) Concerts or instrumental performances using lighting, video, and sound equipment can be held in the park.	☐ **Can currently be held** Regarding adverse impacts of (10) on the environment and organisms: ☐ There are no problems, and no countermeasures have been taken. ☐ Impacts have been mitigated through countermeasures or other adjustments. ☐ No countermeasures have been taken; impacts exist but are tolerated. ☐ No countermeasures have been taken, but the need for improvement is recognized.

If you use any approaches to reduce impacts on the environment or organisms in relation to the items above, please describe them to the extent possible.

Q5. We would like to ask whether measures are being undertaken in this park/green space to conserve the environment while providing enjoyment for park users.

For each of the following items, please check the response that most closely describes the extent to which such measures are currently carried out in practice as part of park policy.

Measures to conserve the environment while providing enjoyment for park users	Please check the option that most closely applies.
(1) Some areas are closed to park users to protect organisms, but opportunities (events) are provided to observe nature in those areas, or arrangements for viewing them from outside (e.g., observation platforms or VR viewing) are provided.	☐ Some form of measure is implemented. ☐ A need is recognized, but the measure is not implemented. ☐ There is no such policy, and the measure is not implemented.
(2) To prevent human activities associated with playground equipment, sports facilities, and eating/drinking facilities from affecting surrounding nature and organisms, measures are taken such as locating facilities away from natural areas, providing visual screening or vegetated buffers, controlling light pollution, and soundproofing.	☐ Some form of measure is implemented. ☐ A need is recognized, but the measure is not implemented. ☐ There is no such policy, and the measure is not implemented.
(3) Hazardous species, such as bees/wasps, spiders, and snakes, are not necessarily all removed; conservation measures are also considered. At the same time, measures to protect park users from these organisms are implemented (e.g., preventing users from entering areas where nests are located, installing protective coverings for users, and providing information about hazardous species). Note: Please answer this item with respect to the removal of organisms that pose a danger to people, rather than with respect to the removal of non-native species addressed in item (8) below.	☐ Some form of measure is implemented. ☐ A need is recognized, but the measure is not implemented. ☐ There is no such policy, and the measure is not implemented.
(4) Regular events involve activities that benefit nature restoration or conservation and are enjoyable experiences in themselves (e.g., pond draining [kaibori], building nest boxes, creating eco-stacks, removing non-native species such as by catching crayfish, and planting and greening).	☐ Some form of measure is implemented. ☐ A need is recognized, but the measure is not implemented. ☐ There is no such policy, and the measure is not implemented.
(5) Park facilities are sometimes built from natural materials out of consideration for organisms, and these facilities are useful to park users (e.g., benches, resting facilities, safety fences, information signs, and interpretive boards).	☐ Some form of measure is implemented. ☐ A need is recognized, but the measure is not implemented. ☐ There is no such policy, and the measure is not implemented.
(6) Conservation activities for organisms are undertaken in the park, but recreational facilities that adversely affect organisms (e.g., sports, rides, and fishing facilities) are not completely excluded; instead, they coexist in separate locations (e.g., fishing is prohibited in some ponds for nature restoration, but is permitted in other ponds instead).	☐ Some form of measure is implemented. ☐ A need is recognized, but the measure is not implemented. ☐ There is no such policy, and the measure is not implemented.
(7) For activities that adversely affect the conservation of organisms in the park, rather than imposing strong restrictions such as prohibitions or penalties, information is provided and users are asked to observe park etiquette, or awareness-raising activities are conducted, encouraging park users to exercise their own judgment.	☐ Some form of measure is implemented. ☐ A need is recognized, but the measure is not implemented. ☐ There is no such policy, and the measure is not implemented.

(8) Organisms that cause problems for nature management, such as non-native species, are not necessarily removed. Efforts include considering safety measures, protecting affected organisms, conducting surveys and monitoring of problematic organisms, and using these activities for environmental education. Note: Please answer this item with respect to removing non-native species to prioritize the conservation of species native to the local area, rather than with respect to removing organisms that pose a danger to people, as addressed in item (3) above.	☐ Some form of measure is implemented. ☐ A need is recognized, but the measure is not implemented. ☐ There is no such policy, and the measure is not implemented.

For the approaches described above to conserve the environment while providing enjoyment for park users, please describe below any specific approaches or independently devised measures.

3. Costs associated with nature restoration

Q6. We would like to ask about funding for nature restoration activities. Please check one applicable option.

(1) Do you ever obtain project funding for the purpose of nature restoration?

☐ Yes ☐ No

(2) Do you allocate funds from the park maintenance budget to cover the costs needed for nature restoration activities?

☐ Yes ☐ No

(3) When undertaking environmental measures such as nature restoration, do you experience budget shortages for any of the following items?

Please check all items for which budget shortages make it difficult to carry out the necessary management.

☐ Development and repair of infrastructure, such as park paths, ponds, and observation spaces, with consideration for nature
☐ Woodland management (e.g., tree planting, pruning, and felling)
☐ Mowing
☐ Installation and repair of noticeboards, access-prevention fences, and similar structures for environmental management
☐ Holding experiential events, such as environmental education activities
☐ Requesting experts or organizations to support activities or provide advice

(4) Regarding nature restoration activities, please describe, to the extent possible, how funds are used, any approaches adopted, and any funding issues (e.g., insufficient funds to carry out nature restoration activities).

4. Complaints and requests

Q7. Please check the following items for which complaints or requests are received (multiple responses permitted).

☐ Complaints/requests concerning vegetation-management issues (e.g., overgrown grass, fallen leaves, fallen trees, and fallen branches).
☐ Complaints/requests concerning wildlife-management issues (reduced convenience for users resulting from prioritizing the conservation of organisms: access restrictions; aversion to or anxiety about organisms; noise, unpleasant odors, etc.).
☐ Complaints/requests concerning the management of recreational facilities (e.g., inadequate facilities).
☐ Complaints/requests concerning event management (e.g., adverse effects of holding events or a limited variety of events).
☐ Complaints/requests concerning safety management (e.g., accidents involving facilities).
☐ Complaints/requests concerning cleanliness (e.g., dumping of trash, or cleanliness of facilities and restrooms).
☐ Complaints/requests concerning staff responses and staffing shortages (e.g., insufficient guidance or support for users).

Q8. How much difficulty do you experience in responding to complaints/requests concerning (1) vegetation management and (2) wildlife management listed in Q7?

For each of the two aspects—the number of complaints/requests and the difficulty of responding—please check the one option that best applies.

(1) Complaints/requests concerning vegetation management (e.g., overgrown grass, fallen leaves, fallen trees, and fallen branches)

Number of complaints/requests (check one)	Difficulty of responding to complaints/requests (check one)
☐ The number of complaints/requests is the highest among the categories listed in Q7.	☐ Most problems cannot be resolved.
☐ The number is not the highest, but there are many complaints/requests.	☐ Approximately half or fewer are resolved.
☐ There are not many, but there are some complaints/requests.	☐ Approximately half or more are resolved.
☐ There are no complaints/requests.	☐ Almost all are resolved.

(2) Complaints/requests concerning wildlife-management issues (reduced convenience for users resulting from prioritizing the conservation of organisms: access restrictions; aversion to or anxiety about organisms; noise, unpleasant odors, etc.)

Number of complaints/requests (check one)	Difficulty of responding to complaints/requests (check one)
☐ The number of complaints/requests is the highest among the categories listed in Q7.	☐ Most problems cannot be resolved.
☐ The number is not the highest, but there are many complaints/requests.	☐ Approximately half or fewer are resolved.
☐ There are not many, but there are some complaints/requests.	☐ Approximately half or more are resolved.
☐ There are no complaints/requests.	☐ Almost all are resolved.

5. Safe recreational use

Q9. We would like to ask about safety measures for park users.

Please check the following measures that are implemented in the park (multiple responses permitted).

☐ Removal of hazardous species that can harm people in the park (bees/wasps and venomous snakes).
☐ Prompt removal of dead or damaged trees and other obstacles from park paths and walkways.
☐ Maintaining clear sightlines through the selection and placement of planted tree species, pruning, and felling.
☐ Installation of lighting in the park to provide a bright and safe environment at night.
☐ Regular inspections of facility safety.
☐ Educational activities on safety measures, such as how to respond to encounters with wildlife.
☐ Regular rounds and patrols to prevent crime.
☐ Producing signs, maps, and similar materials to inform users of hazardous locations.

6. Opinions and impressions

Finally, if you have any general comments on nature restoration and recreational use, please provide them below.

Please freely write any overall impressions of or comments on this questionnaire in the space below.

This completes the questionnaire. Thank you very much for your cooperation.

We hope to use the information and opinions you provide as valuable data for our research.